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Mowing device with a rotary fan and related features

Abstract

An electric motor **15** includes a motor section **15b** and an output shaft **15a** protruding from the motor section **15b**. The output shaft **15a** supports a rotary fan **24**. The rotary fan **24** is drivable by the output shaft **15a** to rotate to supply cooling air to the motor section **15b**.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

(1) This application claims priority to Japanese Patent Application No. 2020-155678 filed Sep. 16, 2020, the disclosure of which is hereby incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(2) The present invention relates to a mowing device.

2. Art Description of Related Art

(3) A mowing device may include a cutting blade housing and cutting blades present inside the cutting blade housing and drivable to rotate. An example of such a mowing device is the mower unit disclosed in Patent Literature 1. The mower unit includes a “mower deck” as the cutting blade housing and “blades” as the cutting blades. A mowing device of this type may include, as disclosed in Patent Literature 1, an electric motor (“work motor” in Patent Literature 1) supported by the cutting blade housing and configured to drive the cutting blades. Patent Literature 1 Japanese Unexamined Patent Application Publication, Tokukai, No. 2019-17278

SUMMARY OF THE INVENTION

(4) The above mowing device will become unable to continue work if the electric motor has been driven and heated up so much as to seize up.

(5) The present invention has an object of providing an inexpensive mowing device configured to drive a cutting blade with use of an electric motor and yet capable of continuing mowing work over an extended period of time.

(6) A mowing device as an embodiment of the present invention includes: a cutting blade housing; a cutting blade present inside the cutting blade housing and drivable to rotate; and an electric motor supported by the cutting blade housing and configured to drive the cutting blade, wherein the electric motor includes: a motor section; and an output shaft protruding from the motor section, and the mowing device further includes a rotary fan supported by the output shaft and drivable by the output shaft to rotate to supply cooling air to the motor section.

(7) The above arrangement allows the rotary fan to be driven by the electric motor to supply cooling air to the motor section and cool the motor section. This prevents the electric motor from becoming heated up excessively, and thereby prevents the electric motor from becoming broken down. The above arrangement, in other words, allows the mowing device to continue mowing work over an extended period of time. Further, the simple structure of the rotary fan being supported by the output shaft allows production of an inexpensive mowing device.

(8) A mowing device as a preferable embodiment of the present invention includes: a fan cover covering the rotary fan and having a vent hole, wherein the rotary fan supplies the cooling air to the motor section by taking in the cooling air from outside the fan cover through the vent hole, and the rotary fan includes a cutter section configured to cut a foreign object having entered the fan cover through the vent hole.

(9) The above arrangement allows the cutter section to cut a foreign object such as mown grass having entered the fan cover. Such a foreign object and/or the like is thus cut to shreds and easily discharged from the fan cover with use of, for example, cooling air. The above arrangement thereby prevents a foreign object and/or the like having entered the fan cover from remaining on, for example, the motor section to prevent the cooling effect from becoming decreased and more

reliably prevent the electric motor from becoming heated up excessively.

(10) A mowing device as a preferable embodiment of the present invention is arranged such that the motor section includes a cooling fin section configured to dissipate heat of the motor section, and the cooling fin section receives the cooling air supplied by the rotary fan.

(11) The above arrangement allows heat of the motor section to be conducted to the cooling fin section and then dissipated by means of the cooling fin section. The above arrangement also promotes dissipation of heat from the cooling fin section due to cooling air that the rotary fan supplies to the cooling fin section. The above arrangement thereby cools the motor section effectively. This more reliably prevents the electric motor from becoming heated up excessively, and thereby effectively prevents the electric motor from becoming broken down.

(12) A mowing device as a preferable embodiment of the present invention is arranged such that the cooling fin section includes a plurality of linear fins present around the output shaft in a plan view.

(13) The above arrangement involves linear fins present over a wide area around the motor section in a plan view. This allows the electric motor to be cooled more effectively through heat dissipation by the cooling fin section.

(14) A mowing device as a preferable embodiment of the present invention is arranged such that the plurality of linear fins extend radially from the output shaft.

(15) The above arrangement allows cooling air supplied by the rotary fan to the cooling fin section to flow along the plurality of linear fins from the output shaft side to the opposite side. This promotes heat dissipation by the cooling fin section, and thereby allows the electric motor to be cooled more effectively.

(16) A mowing device as a preferable embodiment of the present invention includes: an output pulley which is present on the output shaft and between the rotary fan and the motor section and around which a power transmission belt is wound, the power transmission belt interlocking the output shaft and the cutting blade with each other.

(17) The above arrangement allows cooling air supplied from the rotary fan toward the motor section to be brought to the output pulley, meaning that the rotary fan is usable also as a means of cooling the output pulley. This simple structure prevents the output pulley and the power transmission belt from becoming heated up.

(18) A mowing device as a preferable embodiment of the present invention is arranged such that the output pulley and the plurality of linear fins coincide with each other with respect to an axial direction of the output shaft.

(19) The above arrangement allows the rotary fan to be near both the linear fins and the output pulley. This allows the rotary fan to supply cooling air to the linear fins and the output pulley in such a manner as to cool the linear fins and the output pulley effectively.

(20) A mowing device as a preferable embodiment of the present invention is arranged such that the output shaft protrudes upward from the motor section.

(21) The above arrangement allows the rotary fan to be above the motor section. This allows cooling air to be taken in from above the cutting blade housing, and prevents, for example, mown grass from entering the motor section easily.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a left side view of a riding type mower as a whole.

(2) FIG. 2 is a plan view of a mowing device as a whole.

(3) FIG. 3 is a cross-sectional view of the mowing device taken along line III-III in FIG. 2.

(4) FIG. 4 is a cross-sectional view of a motor cooling mechanism for an alternative embodiment.

(5) FIG. 5 is a plan view of a motor cooling mechanism for an alternative embodiment.

(6) FIG. 6 is a graph that shows a change in the temperature of the electric motor of a mowing device which change is caused as the mowing device is driven idly.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(7) The description below deals with an example embodiment of the present invention with reference to drawings.

(8) The embodiment described below is a riding-type mower including a travel vehicle body. FIG. 1 shows arrow F to indicate the forward direction of the vehicle body, arrow B to indicate the backward direction of the vehicle body, arrow U to indicate the upward direction of the vehicle body, and arrow D to indicate the downward direction of the vehicle body. The front side of the drawing corresponds to the leftward direction of the vehicle body, whereas the back side of the drawing corresponds to the rightward direction of the vehicle body.

(9) [Overall Configuration of Riding-Type Mower]

(10) As illustrated in FIG. 1, the riding-type mower includes a travel vehicle body provided with a pair of left and right freely rotatable front wheels 1 and a pair of left and right drivable rear wheels 2. The pair of left and right front wheels 1 are each a caster wheel. The pair of left and right rear wheels 2 are drivable by a pair of left and right travel electric motors 3, respectively. The riding-type mower includes a driver section 4 at a front portion of the travel vehicle body. The driver section 4 includes a driver's seat 5, a driver section floor 6 present in front of and below the driver's seat 5, and a pair of left and right control levers 7 present to the left and right of the driver's seat 5 and operable to move and steer the travel vehicle body. The driver can move and steer the travel vehicle body with use of the left and right control levers 7 by swinging the left and right control levers 7 individually in the front-back direction to change the respective rotation directions and rotation speeds of the left and right travel electric motors 3 by means of respective linkage mechanisms (not shown in the drawings) between the left and right control levers 7 and the left and right travel electric motors 3. The riding-type mower includes a battery pack 8 present at a back portion of the travel vehicle body and configured to supply electric power to, for example, the travel electric motors 3. The riding-type mower includes, between the front wheels 1 and the rear wheels 2, a mowing device 9 configured to mow turf and grass and discharge mown turf and grass (hereinafter referred to simply as "mown grass") laterally outward of the vehicle body. The mowing device 9 is supported by a vehicle body frame 10 via a link mechanism 11 configured to allow the mowing device 9 to be lifted and lowered relative to the vehicle body frame 10.

(11) [Configuration of Mowing Device]

(12) FIG. 2 is a plan view of the mowing device 9 as a whole. The description below of the mowing device 9 refers to FIG. 2, which shows arrow F1 to indicate the forward direction of the device, arrow B1 to indicate the backward direction of the device, arrow L1 to indicate the leftward direction of the device, and arrow R1 to indicate the rightward direction of the device. The mowing device 9 is oriented so that its front-back direction is identical to the front-back direction of the travel vehicle body.

(13) As illustrated in FIG. 2, the mowing device 9 includes a cutting blade housing 12 and two cutting blades 13 present inside the cutting blade housing 12 and drivable to rotate. The cutting blades 13 are arranged next to each other in the left-right direction of the device. The cutting blades 13 are each in the shape of a blade, and are each supported by the cutting blade housing 12 via a rotary shaft 13a extending in the up-down direction of the device. The mowing device 9 includes, to the right of the cutting blade housing 12, a discharge guide 14 connected to the cutting blade housing 12 and configured to guide mown grass discharged from the cutting blade housing 12 toward a lateral side of the vehicle body. The mowing device 9 includes a motor device M supported by a back portion of the cutting blade housing 12 and including an electric motor 15 configured to drive the cutting blades 13. The mowing device 9 includes, on an upper surface side of the cutting blade housing 12, (i) a power transmission belt 16 configured to transmit power from the electric motor 15 to the cutting blades 13 and (ii) a power transmission section cover 17

covering, for example, the power transmission belt **16**. The power transmission section cover **17** includes a back portion **17a** covering the motor device **M**.

(14) [Configuration of Motor Device]

(15) As illustrated in FIGS. **2** and **3**, the motor device **M** includes an electric motor **15** present under the back portion **17a** of the power transmission section cover **17** and configured to drive the cutting blades **13**. The electric motor **15** includes a motor section **15b** including, for example, a rotor and an output shaft **15a** protruding from the motor section **15b**. The motor section **15b** is, as illustrated in FIG. **2**, in the shape of a circle with the output shaft **15a** as its center in a plan view. The output shaft **15a**, as illustrated in FIG. **3**, protrudes upward from the motor section **15b**. The motor section **15b** includes a front portion between (i) an arc-shaped back portion of that part of the cutting blade housing **12** which contains the left cutting blade **13** and (ii) an arc-shaped back portion of that part of the cutting blade housing **12** which contains the right cutting blade **13**.

(16) The electric motor **15** is attached to a motor support **18** extending backward from a back portion of the cutting blade housing **12**. The electric motor **15** is thereby supported by the cutting blade housing **12** via the motor support **18**. The motor support **18**, as illustrated in FIGS. **2** and **3**, includes (i) a motor mount section **18a** which extends backward from the cutting blade housing **12** and on which the motor section **15b** is mounted, (ii) a connecting bar section **18b** connecting a back portion of the motor mount section **18a** to the cutting blade housing **12**, and (iii) an upper support section **18c** extending from the motor mount section **18a** to a height above the motor section **15b** and supporting an upper portion of the motor section **15b** from above.

(17) As illustrated in FIGS. **2** and **3**, the motor device **M** includes an output pulley **20** on the output shaft **15a**. The motor device **M** includes a power transmission belt **16** wound around the output pulley **20** and cutting blade drive pulleys **21** each present on the rotary shaft **13a** of the corresponding cutting blade **13**. The cutting blade drive pulleys **21** are each present on that portion of the corresponding rotary shaft **13a** which protrudes upward from the cutting blade housing **12**. As illustrated in FIG. **2**, the power transmission belt **16** is wound around a guide wheel **22** between the left cutting blade drive pulley **21** and the output pulley **20**. The guide wheel **22** is supported by the cutting blade housing **12**.

(18) The motor device **M** is configured such that the electric motor **15** transmits power from the output shaft **15a** to each cutting blade drive pulley **21** via the power transmission belt **16** and that each cutting blade drive pulley **21** then transmits power to the corresponding cutting blade **13** via its rotary shaft **13a**, whereby the electric motor **15** drives the cutting blades **13**.

(19) [Configuration of Motor Cooling Mechanism]

(20) The motor device **M** is, as illustrated in FIG. **3**, provided with a motor cooling mechanism **23** configured to cool the electric motor **15** so that the electric motor **15** is drivable continuously over an extended period of time.

(21) The motor cooling mechanism **23**, as illustrated in FIG. **3**, includes a rotary fan **24** configured to cool the electric motor **15**. The rotary fan **24** is supported by the output shaft **15a** in such a manner as to be drivable by the output shaft **15a** to rotate. Specifically, as illustrated in FIG. **3**, the rotary fan **24** includes a fan section **24a** and a connecting section **24b** inside the fan section **24a**. The fan section **24a** includes blower fins **24f** arranged apart from each other along the circumference of the rotary fan **24**. The connecting section **24b** has an attachment hole **24c** at a central portion thereof. The output pulley **20** is provided with a fan support section **20a** protruding upward from a side portion of the output pulley **20**. The connecting section **24b** is fitted around an upper end portion **20b** of the fan support section **20a** with the upper end portion **20b** in the attachment hole **24c**. The connecting section **24b** includes a depressed portion **24d** in an inner circumferential portion thereof, whereas the upper end portion **20b** includes a protruding portion **20c** fitted in the depressed portion **24d**. The connecting section **24b** and the fan support section **20a** are thereby engaged with each other in such a manner as to be incapable of rotation relative to each other. The rotary fan **24** is supported by the output shaft **15a** via the output pulley **20**. The rotary

fan **24** and the output shaft **15a** are connected to each other via the output pulley **20** in an interlocking manner, so that the output shaft **15a** is capable of driving the rotary fan **24** to rotate. (22) As illustrated in FIG. 3, the output pulley **20** is present on the output shaft **15a** between the rotary fan **24** and the motor section **15b**.

(23) As illustrated in FIGS. 2 and 3, the motor device M includes a fan cover **25** that surrounds and covers the rotary fan **24**. The fan cover **25** protrudes upward from the back portion **17a** of the power transmission section cover **17**. The fan cover **25** includes a top plate portion **25a** having a plurality of vent holes **26** that allow the fan cover **25** to take in cooling air from outside. The vent holes **26** are each in the shape of a slit for the present embodiment, but are not limited as such. The vent holes **26** may alternatively each have any of various shapes such as a circle, a rectangle, or an ellipse. As illustrated in FIGS. 2 and 3, the power transmission section cover **17** has, at an edge portion thereof, a plurality of exhaust ports **32** that allow air from the rotary fan **24** to be discharged.

(24) As illustrated in FIG. 3, the rotary fan **24** includes a cutter section **27** configured to cut a foreign object such as mown grass having entered the fan cover **25**. The cutter section **27** is in the form of a circular screen, and is present at an upper end portion of the rotary fan **24** in such a manner as to face the top plate portion **25a** of the fan cover **25**. The cutter section **27** has a plurality of through holes **27a**. The cutter section **27** is provided with a connecting cylindrical portion **28** extending from a central portion of the cutter section **27** toward the fan support section **20a**. The connecting cylindrical portion **28** includes, at a lower end portion thereof, a flange portion **28a** placed on the connecting section **24b** of the rotary fan **24** and fastened to the fan support section **20a** with use of a connecting bolt. The cutter section **27** is thus supported by the rotary fan **24** in such a manner as to be incapable of rotation relative to the rotary fan **24**. The flange portion **28a** serves to prevent the rotary fan **24** from becoming detached from the fan support section **20a**. The through holes **27a** are each in the shape of a circle for the present embodiment, but are not limited as such. The through holes **27a** may alternatively each have any of various shapes such as a slit, a rectangle, or an ellipse. If a foreign object enters the fan cover **25** through a vent hole **26** to have one end in a through hole **27a** and the other end in the vent hole **26**, the foreign object is caught by the top plate portion **25a** and the cutter section **27**. In this state, the cutter section **27** is rotated relative to the top plate portion **25a** to cut the foreign object.

(25) The motor cooling mechanism **23** is arranged such that the rotary fan **24** is drivable by the electric motor **15** to rotate about the axis Y of the output shaft **15a**. The rotation of the rotary fan **24** takes in cooling air from outside the fan cover **25** through the vent holes **26**. The cooling air flows from an inner portion of the rotary fan **24** through a space between the blower fins **24f** to a space around the rotary fan **24**. The cooling air is then partly supplied to the output pulley **20** to cool the output pulley **20** and the power transmission belt **16**. The cooling air having flown to the space around the rotary fan **24** is also partly supplied to the motor section **15b** through the upper support section **18c** or directly to the motor section **15b** past a space in which the upper support section **18c** is absent. The cooling air thereby cools the motor section **15b**. Cooling air having entered the power transmission section cover **17** flows through the exhaust ports **32** or openings **33** in the cutting blade housing **12** to outside the power transmission section cover **17**. If the motor device M has taken cooling air into the fan cover **25** together with a foreign object such as mown grass through a vent hole **26**, the cutter section **27** shreds the foreign object. The motor device M discharges the shredded foreign object together with cooling air.

(26) [Alternative Embodiments]

(27) (1) FIG. 4 is a cross-sectional view of a motor cooling mechanism **30** for an alternative embodiment. The motor cooling mechanism **30** for an alternative embodiment includes a rotary fan **24**, and also includes a cooling fin section **31** for the motor section **15b**.

(28) As illustrated in FIGS. 4 and 5, the cooling fin section **31** receives heat from the motor section **15b** through heat transfer involving contact with the motor section **15b** and heat transfer involving

no contact with the motor section **15b**. The cooling fin section **31** also dissipates heat that it has received. The cooling fin section **31** is oriented to face the rotary fan **24** to receive cooling air from the rotary fan **24**.

(29) The cooling fin section **31** is mounted over the motor section **15b**. The upper support section **18c** is connected to the cooling fin section **31** and the motor mount section **18a**. The cooling fin section **31** is pressed by the upper support section **18c** against the motor section **15b**, so that the cooling fin section **31** serves to fix the electric motor **15** to the motor mount section **18a**.

(30) The cooling fin section **31**, as illustrated in FIGS. **4** and **5**, includes a mount section **31a** placed on and connected to the motor section **15b** and a plurality of linear fins **31b** protruding upward from the mount section **31a**. The linear fins **31b** are each, for example, a plate-shaped fin standing on the mount section **31a**. The linear fins **31b** each have a predetermined height and a predetermined thickness in a plan view. The linear fins **31b** are, however, not limited as such, and may each be a member having any shape with a predetermined surface area and supported by the mount section **31a**. The plurality of linear fins **31b** are, as illustrated in FIG. **5**, arranged around the output shaft **15a** in a pattern similar to a radial pattern with the output shaft **15a** as the center. The mount section **31a** has, at a portion forward of the motor section **15b**, a through hole **29** that allows air to pass therethrough. The through hole **29** is, as illustrated in FIG. **5**, in the shape of a substantial triangle in a plan view. As illustrated in FIG. **4**, those linear fins **31b** which are under the power transmission belt **16** each protrude from the mount section **31a** to a height smaller than the height to which each of the other linear fins **31b** protrudes from the mount section **31a**.

(31) As illustrated in FIG. **4**, the output pulley **20** and the plurality of linear fins **31b** coincide with each other with respect to the axial direction of the output shaft **15a**. This allows the rotary fan **24** to be near both the linear fins **31b** and the output pulley **20**.

(32) The motor cooling mechanism **30** is configured such that heat generated by the motor section **15b** is conducted to the cooling fin section **31** and is then dissipated by the linear fins **31b**, whereby the cooling fin section **31** cools the electric motor **15**. The rotary fan **24** takes in cooling air from outside the fan cover **25**. The cooling air flows through a space around the rotary fan **24**, and is supplied to the cooling fin section **31**. The cooling air then flows along the plurality of linear fins **31b** from the output shaft side (that is, the side close to the output shaft **15a**) toward the opposite side (that is, the side far from the output shaft **15a**). This allows the cooling air to cool the linear fins **31b** and promote heat dissipation by the linear fins **31b**. The cooling air having flown along the linear fins **31b** toward the opposite side flows through the exhaust ports **32** and the openings **33** to be discharged from the power transmission section cover **17**.

(33) FIG. **6** is a graph that shows the result of a test on a change in the temperature of the electric motor **15** of a mowing device **9** including the cooling fin section **31** and the rotary fan **24** which change is caused as the mowing device **9** is driven idly. The graph has (i) a horizontal axis indicative of the time length over which the mowing device **9** is driven idly and (ii) a vertical axis indicative of the increase in the temperature of the electric motor **15**. The temperature increase is a difference between the temperature of the electric motor **15** and that of outside air. The graph shows (i) circles to indicate a temperature change for a case involving no cooling fin section **31** or rotary fan **24** and (ii) triangles to indicate a temperature change for a case involving the cooling fin section **31** and the rotary fan **24**.

(34) (2) The embodiment described above is arranged such that the output shaft **15a** protrudes upward from the motor section **15b** and that the output pulley **20** and the rotary fan **24** are present above the motor section **15b**. The embodiment may, however, be altered such that the output shaft **15a** protrudes downward from the motor section **15b** and that the output pulley **20** and the rotary fan **24** are present below the motor section **15b**.

(35) (3) The embodiment described above is an example in which the linear fins **31b** are arranged in a pattern similar to a radial pattern with the output shaft **15a** as the center. The present invention is, however, not limited to such an arrangement. The embodiment may be altered to include, for

example, (i) linear fins extending in the left-right direction of the cutting blade housing **12** and arranged next to one another in the front-back direction of the cutting blade housing **12** or (ii) linear fins extending in the front-back direction of the cutting blade housing **12** and arranged next to another in the left-right direction of the cutting blade housing **12**.

(36) (4) The embodiment described above is an example in which the output pulley **20** and the linear fins **31b** coincide with each other with respect to the axial direction of the output shaft **15a**. The present invention is, however, not limited to such an arrangement. The embodiment may be altered, for instance, such that the linear fins **31b** are above or below the output pulley **20**.

INDUSTRIAL APPLICABILITY

(37) The present invention is applicable to a mowing device including a cutting blade housing, cutting blades present inside the cutting blade housing and drivable to rotate, and an electric motor supported by the cutting blade housing and configured to drive the cutting blades.

REFERENCE SIGNS LIST

(38) **12** Cutting blade housing **13** Cutting blade **15** Electric motor **15a** Output shaft **15b** Motor section **16** Power transmission belt **20** Output pulley **24** Rotary fan **25** Fan cover **26** Vent hole **27** Cutter section **31** Cooling fin section **31b** Linear fin

Claims

1. A mowing device, comprising: a cutting blade housing; a cutting blade present inside the cutting blade housing and drivable to rotate; and an electric motor supported by the cutting blade housing and configured to drive the cutting blade, wherein the electric motor comprises: a motor section; and an output shaft protruding from the motor section, and wherein the mowing device further comprises: a rotary fan supported by the output shaft and drivable by the output shaft to rotate to supply cooling air to the motor section; a fan cover covering the rotary fan; at least one vent hole formed in the fan cover; and a circular cutter section disposed between the rotary fan and the fan cover, connected to the rotary fan, supported by the rotary fan in such a manner as to be incapable of rotation relative to the rotary fan, rotated relative to the fan cover, and centered on an axis of the output shaft; wherein the cutter section comprises a circular screen and at least one circular through hole provided in the circular screen, wherein the rotary fan supplies the cooling air to the motor section by taking in the cooling air from outside the fan cover through the at least one vent hole, and wherein the cutter section is rotated relative to the fan cover to cut a foreign object, one end of the foreign object having entered the circular through hole and another end of the foreign object existing in the at least one vent hole.
 2. The mowing device according to claim 1, wherein: the motor section comprises a cooling fin section configured to dissipate heat of the motor section, and the cooling fin section receives cooling air supplied by the rotary fan.
 3. The mowing device according to claim 2, wherein the cooling fin section comprises a plurality of linear fins present around the output shaft in a plan view.
 4. The mowing device according to claim 3, wherein the plurality of linear fins extend radially from the output shaft.
 5. The mowing device according to claim 3, further comprising: an output pulley which is present on the output shaft and between the rotary fan and the motor section and around which a power transmission belt is wound, the power transmission belt interlocking the output shaft and the cutting blade with each other.
 6. The mowing device according to claim 5, wherein the output pulley and the plurality of linear fins coincide with each other with respect to an axial direction of the output shaft.
 7. The mowing device according to claim 1, wherein the output shaft protrudes upward from the motor section.
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